

Rudra Amin

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Education

Northern Arizona University Flagstaff, AZ

August 2021 – May 2025

Bachelor's in computer engineering

Relevant Coursework: **Python, C, C++, Data structures & Algorithms**, Programmed microprocessor using **SystemVerilog** for hardware design and verification, **Machine Learning, Natural Language Processing, Data Science, Full-Stack Web Development.**

Skills

Software: Python, C, C++, Java, HTML, CSS, SQL, SystemVerilog.

Deep Learning: Neural Networks, CNN, RNN, Transfer learning, Time-Series Analysis, NLP, Tokenization, Embeddings, Sequence Modeling, NER, Text Classification, Signal Classification, Object Detection, Feature Extraction, Image Augmentation & Classification.

Tools & DevOps: REST APIs, Node, React, Flask Git, AWS, Docker, MongoDB, Kubernetes, Jenkins (CI/CD), Bootstrap.

Soft Skills: Strong problem-solving, leadership, and collaboration skills; adept in Excel, Word, and all MS 365 applications, with the ability to work independently or as part of a team. Hand on experience with Windows and MacOS systems.

Work Experience

IT Student Technician Northern Arizona University

April 2022 – Present

- Delivered front-line technical support across Windows and MacOS platforms resolving hardware, software, and network issues for students, faculty, and staff (including Microsoft Office 365, Adobe Creative Suite, Louie/PeopleSoft, Canvas, Zoom, etc.) while maintaining a 90%+ customer satisfaction rate over 9,000+ support tickets.
- Led reimaging and setup of university-owned desktops and laptops, configuring user environments, installing necessary drivers, updating operating systems, and ensuring secure, up-to-date workstations.
- Trained new technicians on standard operating procedures, ticketing systems (**ServiceNow**), and customer interaction best practices.

Software Developer Intern DSD Lab

August 2024 – December 2024

- Developed and tested **Java** and **Swift** (iOS) applications, contributing to the completion of 3 major project milestones within the internship period, while improving application performance by 20% through code optimization using **data structures and algorithms** as well as **effective debugging**.
- Developed a full-stack web dashboard using **React, HTML, CSS, and Bootstrap** to visualize real-time application metrics and ML-driven anomaly alerts.
- Designed, trained, and deployed a machine learning anomaly-detection model in Python (scikit-learn) to flag potential application errors from usage logs; packaged the model as a Flask microservice, reducing incident response time by 30%.
- Containerized microservices using **Docker** and orchestrated deployments on **AWS EKS (Kubernetes)**, ensuring zero-downtime rollouts and horizontal scalability under peak load.
- Wrote unit and integration tests using **JUnit** and **XCTest** to ensure code reliability and minimize bugs during deployment.
- Authored **SQL** queries and stored procedures to optimize relational data retrieval and collaborated on database schema design in **PostgreSQL** to support new feature requirements.

Projects

1. Dietary Classification:

- Developed an feature extraction methodology using **image classification**, increasing ingredient categorization accuracy by 20% and streamlining dataset preparation for machine learning models. (More info in GitHub Repository)
- Integrated Google Translate API to ensure consistent data representation across multiple languages, enhancing the product's international applicability.

2. SkimLit:

- Preprocessed a 20K-abstract PubMed RCT dataset by writing custom **tokenization/segmentation** scripts in **Python** (using Pandas, NumPy, and built-in string methods) to split each abstract into labeled sentences (Objective, Methods, Results, etc.), and encoded labels with Scikit-learn's LabelEncoder and OneHotEncoder.
- Benchmarked a traditional text-classification baseline using TF-IDF vectorization + Multinomial Naive Bayes in a sklearn Pipeline, achieving a ~85% accuracy on five abstract-section labels.
- Packaged the final inference pipeline so that given a raw abstract, the model automatically tags each sentence with its section label accelerating medical literature review and cutting manual annotation time for students by 40%.

3. StudySphere:

- Co-developed with a team of 6 developers; A **MERN stack** application that simplifies creating, joining, and managing study groups, connecting students with like-minded peers to "study smarter together."
- utilizing **MongoDB** for efficient data storage and management of user profiles, groups, and messages.
- Implemented features such as **real-time messaging** and optimized scheduling to support virtual and in-person collaboration.
- Collaborated on this group project over the course of one semester, contributing to design, development, and deployment efforts.

4. Motion Sensor Alarm Clock:

- Co-developed an **Arduino**-based alarm clock system with an **embedded motion sensor** to detect user presence in bed, designed to drop water as a wake-up mechanism if motion is detected in bed after the alarm goes off.
- Authored the Arduino firmware in **C/C++** to continuously poll the motion sensor for period set by the user/consumer, implementing signal filtering (debounce logic and moving-average smoothing) to eliminate false positives and ensure accurate bed-presence detection.
- Collaborated with a team of 3 peers on hardware integration, motion detection, and system functionality.

Achievements & Leadership

- 4x dean's list** (avg GPA above 3.5 for the semester)
- Led a year-long capstone team developing an AI-driven ECG classification system, coordinating all research and development tasks, mentoring teammates.
 - 1st place in the Electrical Engineering category at the NAU Engineering Fest.**